

6.2

$$u(y) = Ay + By^2 - Cy^3$$

$$T(y) = D + Ey + Fy^2 - Gy^3$$

$$C_f = \frac{\tau_w}{\rho u_{\infty}^2 / 2} \quad \text{3 점의}$$

$$= \frac{N \frac{du}{dy} \big|_{y=0}}{\rho u_{\infty}^2 / 2}$$

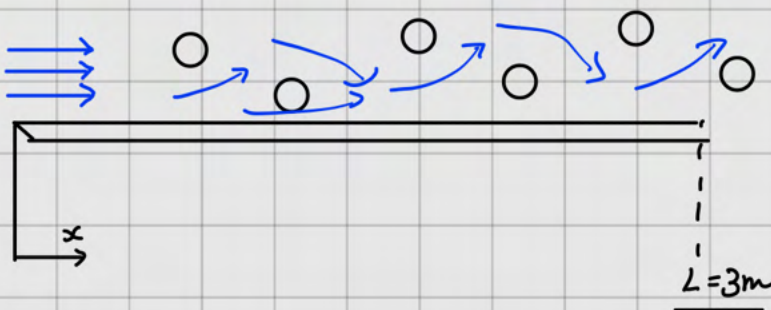
$$= \frac{2AN}{\rho u_{\infty}^2}$$

$$\xi_s'' = -k_2 \frac{dT}{dy} \big|_{y=0} = h(T_s - T_{\infty})$$

$$\therefore h = \frac{-k_2 \frac{dT}{dy} \big|_{y=0}}{T_s - T_{\infty}}$$

$$= \frac{-k_2 E}{D - T_{\infty}} \quad (T_s = T(0) = D)$$

6.6.



$$h_x = 0.7 + 3.6x - 3.4x^2$$

$$\bar{h}_L = \frac{1}{L} \int_0^L h_x dx$$

$$= \frac{1}{L} \int_0^L (0.7 + 3.6x - 3.4x^2) dx$$

$$= \frac{1}{L} \times \left[0.7x + 1.8x^2 - \frac{3.4}{3}x^3 \right]_0^L$$

$$= 0.7 + 1.8L - \frac{3.4}{3}L^2$$

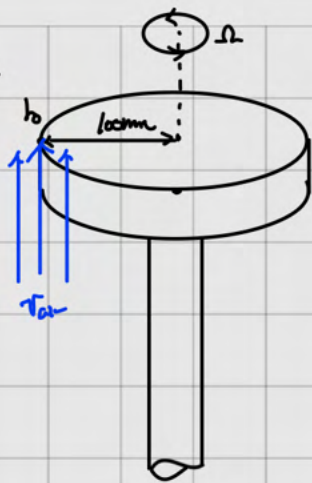
$$= 10.9 \text{ W/m}^2\cdot\text{K}$$

$$h_L = 0.7 + 3.6L - 3.4L^2$$

$$= 10.9 \text{ W/m}^2\cdot\text{K}$$

$$\therefore \frac{\bar{h}_L}{h_L} = 1$$

6.14.



$$h = 20 \text{ W/m}^2 \cdot \text{K}$$

$$T_{\infty} = 20^\circ\text{C}$$

$$T_s = 50^\circ\text{C}$$

local heat flux?

total electric power?

Steady State.

disk를 잘라내었고 공기가 흐르는 것으로 생각할 수 있다.

한 걸어서 $h = 20 \text{ W/m}^2 \cdot \text{K}$

$$\therefore q'' = h(T_s - T_{\infty}) = 20 \times (50 - 20) = \underline{600 \text{ W/m}^2}$$

disk에 C.V.를 설정.

$$\frac{dE_{cv}}{dt} = \dot{Q} - \dot{W}$$

$$\dot{Q} = \dot{W} = P_{elec}$$

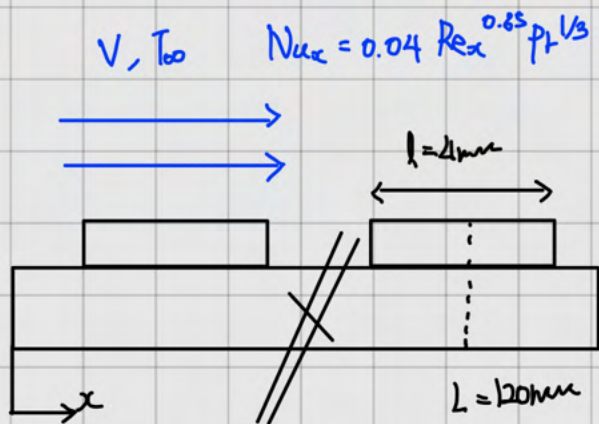
$$\therefore \dot{Q} A = P_{elec}$$

$$P_{elec} = (600 \text{ W/m}^2) \times \pi (0.1)^2$$

$$= \underline{18.8 \text{ W}}$$

속도분포가 온도분포에 영향을 주고 온도분포가 h 값의 결정에 기여하는데 h 가 너무 복잡하게 (아래 이쁘고 boundary layer의 두께는 변할 것이다. (외복류층에서도 더이상 밝히려지 않음). (속도분포와 연결)

6.33



$$T_\infty = 25^\circ\text{C}$$

$$V = 10 \text{ m/s}$$

4 mm x 4 mm chip

$$T_s \text{ at } P = 30 \text{ mW}$$

칩이 놓여 있지. Steady state, 칩은 작아서 1cm 이하의 평균치는 같다.

chip에 C.V. 설정.

$$\frac{dE}{dt} = \dot{Q} - \dot{W}$$

$$\dot{E}_{\text{conv}} = \dot{W} = P$$

$$\therefore \dot{E}_{\text{conv}} = 30 \text{ mW} \quad \dots (1)$$

$$\text{at } x = 20 \text{ mm}$$

$$Nu_x = 0.04 \left(\frac{Vx}{\nu} \right)^{0.85} \left(\frac{\nu}{\alpha} \right)^{1/3}$$

$$\frac{hL}{k} = 498.948 \dots$$

$$h = 87.5 \text{ W/m}^2\cdot\text{K} = \bar{h}$$

$$\therefore \dot{E}_{\text{conv}} = \bar{h} A (T_s - T_\infty)$$

$$30 \times 10^{-3} = (87.5) (16 \times 10^{-6}) (T_s - 25)$$

$$\therefore \underline{T_s = 46.4^\circ\text{C}} \quad - \text{property 라이브러리를 통해 } 9.17\% \text{ 발생}$$

Table 1.4, at $P = \text{water}$ $T = 300 \text{ K}$

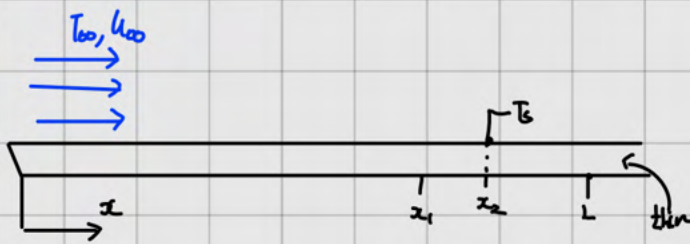
$$c_p = 1.007 \text{ kJ/kg}\cdot\text{K}$$

$$\nu = 15.89 \times 10^{-6} \text{ m}^2/\text{s}$$

$$\alpha = 22.5 \times 10^{-6} \text{ m}^2/\text{s}$$

$$k = 26.3 \times 10^{-3} \text{ W/m}\cdot\text{K}$$

11.6



Provide three expression about $\bar{h}_{1-2}$

$$h_{x_1} > h_{x_2} \quad \therefore \left. \frac{\partial u}{\partial y} \right|_{x=x_1} > \left. \frac{\partial u}{\partial y} \right|_{x=x_2}$$

a) h at $\bar{x} = \frac{x_1 + x_2}{2}$

x_1, x_2 사이 간격이 충분히 작으면 $x_1 \approx \frac{x_1 + x_2}{2} \approx x_2$

$$\therefore \bar{h}_{1-2} = h_{\bar{x}}$$

b) h at x_1, x_2

x_1, x_2 사이 간격이 충분히 작으면 h 가 연속적으로 변화하므로

$$\therefore \bar{h}_{1-2} = \frac{h_{x_1} + h_{x_2}}{2}$$

c) Average coefficients At x_1, x_2

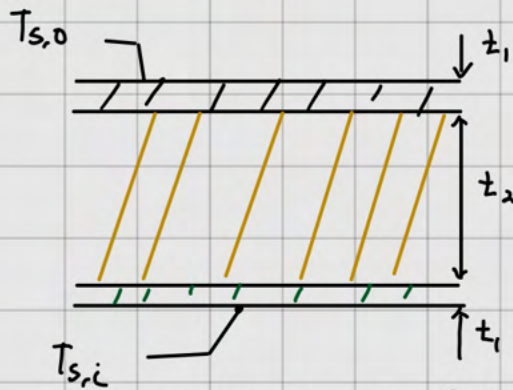
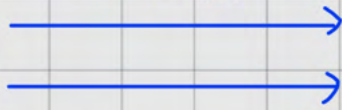
조건을 나누면

$$\delta_{1-2} = \delta_{0-2} - \delta_{0-1}$$

$$\bar{h}_{1-2} (x_2 - x_1) (T - T_{\infty}) = \bar{h}_{0-2} x_2 (T - T_{\infty}) - \bar{h}_{0-1} x_1 (T - T_{\infty})$$

$$\bar{h}_{1-2} = \frac{x_2}{x_2 - x_1} \bar{h}_{0-2} - \frac{x_1}{x_2 - x_1} \bar{h}_{0-1}$$

11.16. $V = 105 \text{ km/h} = 29.16 \text{ m/s}$, $T_{\infty} = 32^\circ\text{C}$, turbulent



$$t_1 = 5 \text{ mm} \quad k_p = 180 \text{ W/m}\cdot\text{K}$$

$$t_2 = 50 \text{ mm} \quad k_i = 0.026 \text{ W/m}\cdot\text{K}$$

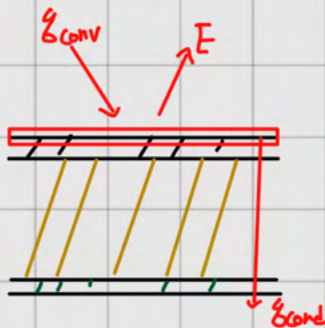
$$T_{s,i} = -10^\circ\text{C}$$

$$G_3 = 1750 \text{ W/m}^2 \quad \alpha_s = \epsilon = 0.5$$

$$A = 10 \text{ m} \times 3.5 \text{ m} = 35 \text{ m}^2$$

1-D heat transfer, steady state, plate 722, $h = \text{const}$

a) $T_{s,o}$? Corresponding heat load?



다음과 같이 C.V.를 설정한다.

$$\frac{dE}{dt} = \dot{Q} - \dot{W}$$

$$\dot{q}_{\text{conv}} + \alpha_s G_3 = \dot{q}_{\text{cond}} + E$$

$$h(T_{\infty} - T_{s,o}) + \alpha_s G_3 = \frac{T_{s,o} - T_{s,i}}{2 \times \frac{t_1}{k_p} + \frac{t_2}{k_i}} + E \sigma T_{s,o}^4$$

대입하면

$$T_{s,o} = 34.7^\circ\text{C} = 312.85 \text{ K}$$

refrigeration system 으로 들어온 heat = $\dot{q}_{\text{cond}}$

$$\therefore \dot{q}_{\text{cond}} = \frac{T_{s,o} - T_{s,i}}{2 \times \frac{t_1}{k_p} + \frac{t_2}{k_i}} \times A = 904.6 \text{ W}$$

$$P = 1 \text{ atm} \quad T = 300 \text{ K}$$

for turbulent flow over isothermal plate

$$Nu_x = 0.0296 Re^{4/5} Pr^{1/3}$$

$$= 0.0296 \left(\frac{V L}{\nu} \right)^{4/5} (0.707)^{1/3}$$

$$\bar{h} = \frac{k}{L} \times 0.0296 \left(\frac{V L}{\nu} \right)^{4/5} (0.707)^{1/3}$$

$$= 44.96 \text{ W/m}^2\cdot\text{K}$$

b) $\alpha_s = 0.15$, $\epsilon = 0.8$ $T_{s,o}$? Corresponding heat load?

위의 식에 값을 대입하면

$$T_{s,o} = 32.0^\circ\text{C} = 305.15 \text{ K}$$

$$\dot{q} = 1764.4 \text{ W}$$

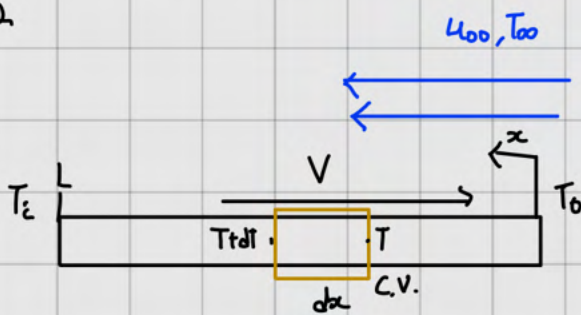
c) $\alpha_s = \epsilon = 0.5$, $t_2 = 0$ $T_{s,o}$? Corresponding heat load?

위의 식에 값을 대입하면

$$T_{s,o} = -9.87^\circ\text{C} = 263.28 \text{ K}$$

$$\dot{q} = 81834.5 \text{ W}$$

7.22

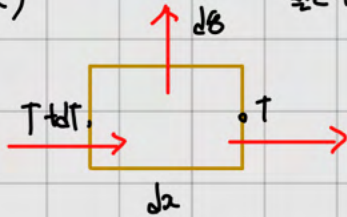


$w, \beta, V, \rho, c_p, h$

u_{∞}, T_{∞}

Steady state, constant property

a) $\dot{Q}$ 가 일정하며 에너지가 이동한다.



C.V.에서

$$\dot{m} c_p (T + dT) = d\dot{Q} + \dot{m} c_p T$$

$$d\dot{Q} = \dot{m} c_p dT$$

$$h_x W (T - T_{\infty}) dx = \int A V c_p dT$$

$$\frac{dT}{dx} = \frac{h_x}{\int V \beta c_p} (T - T_{\infty})$$

2024-16

$$\rho = 2780 \text{ kg/m}^3$$

$$c_p = 875 \text{ J/kg}\cdot\text{K}$$

$$\text{let } T - T_{\infty} = \theta$$

$$\frac{dT}{dx} = \frac{d\theta}{dx}$$

$$\therefore \frac{d\theta}{dx} = \frac{h_x}{\int V \beta c_p} \theta$$

$$\int_{\theta_i}^{\theta_o} \frac{d\theta}{\theta} = \int_0^L \frac{h_x}{\int V \beta c_p} dx$$

$$\ln \frac{\theta_o}{\theta_i} = \frac{L h_x}{\int V \beta c_p}$$

$$\frac{T_i - T_{\infty}}{T_o - T_{\infty}} = \exp\left(\frac{L h_x}{\int V \beta c_p}\right)$$

$$\therefore T_o = \exp\left(-\frac{L h_x}{\int V \beta c_p}\right) (T_i - T_{\infty}) + T_{\infty}$$

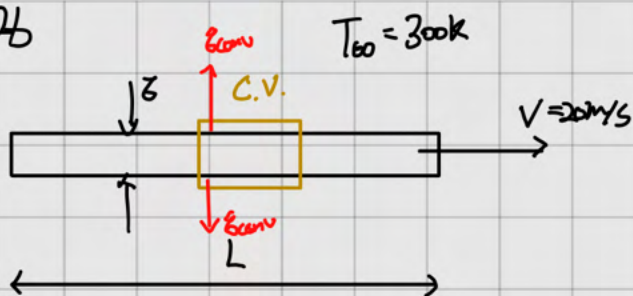
$$b) \text{Re}_L = \frac{u_{\infty} L}{\nu} = 4.31 \times 10^6 \text{ (Turbulent)}$$

$$\therefore \text{Nu}_x = 0.0296 \text{Re}^{4/5} \text{Pr}^{1/3}$$

$$\therefore \bar{h}_L = 37.4 \text{ W/m}^2\cdot\text{K}$$

$$\therefore T_o = 185.7^\circ\text{C}$$

17.26



$$D = 0.002 \text{ m}$$

$$L = 175 \text{ m}$$

$$C_p = 640 \text{ J/kg}\cdot\text{K}$$

$$\rho = 11900 \text{ kg/m}^3$$

No radiation, conduction $\rightarrow$ lumped system

$$\text{C.V. on } k$$

$$\frac{dT}{dt} = \dot{Q} - \dot{W}$$

$$= \dot{Q}_{\text{conv}}$$

$$\rho V C_p \frac{dT}{dt} = 2 h_x A (T - T_\infty)$$

$$\rho C_p \frac{dT}{dt} = 2 h_x (T - T_\infty)$$

$$\therefore \frac{dT}{dt} = \frac{2 h_x}{\rho C_p} (T - T_\infty)$$

$$\text{At } x = 0.1 \text{ m}$$

$$Re = \frac{Vx}{\nu} = 2.41 \times 10^4 \quad \therefore \text{laminar}$$

$$\therefore Nu_x = 0.332 Re^{1/2} Pr^{1/3}$$

$$h = \frac{k}{x} \cdot 0.332 Re^{1/2} Pr^{1/3} \quad Pr = 0.7$$

$$= 25.1 \text{ W/m}^2\cdot\text{K}$$

$$\therefore \left. \frac{dT}{dt} \right|_{x=0.1 \text{ m}} = -4.46$$

$$\text{At } x = 175 \text{ m}$$

$$Re = \frac{Vx}{\nu} = 1.94 \times 10^7 \quad (\text{turbulent})$$

$$\therefore Nu_x = 0.6296 Re^{4/5} Pr^{1/3}$$

$$\therefore h = \frac{k}{x} \cdot 0.6296 Re^{4/5} Pr^{1/3} = 13 \text{ W/m}^2\cdot\text{K}$$

$$\therefore \left. \frac{dT}{dt} \right|_{x=175 \text{ m}} = -2.31$$

transition 발생할 때 h 의 값의 차이

$$Re_x = 5 \times 10^5 \rightarrow x = 1.91 \text{ m}$$

